
Predicting Probability of Default for Banca Massiccia

Business Understanding

- Banca Massiccia, a large Italian bank, has been making loans to businesses for many years, and has experimented with different approaches to underwriting these loans, including statistical models of default. Banca Massiccia is interested in understanding how to reduce its portfolio default rate.
- The bank aims to be able to better predict the probability that a potential borrower will default on a principal or interest payment for a prospective loan over the next 12 months.
- Our approach ingests raw financial data and firm-specific attributes to predict the probability of defaulting within the next 12 months, based on financially grounded indicators. The predicted PD can then be used by the bank for pricing and underwriting decisions.



Problem Formulation

- Each row represents a borrower at a given firm year, with their financial attributes recorded at the statement date.
- We begin by examining the default date (def_date) associated with each firm.
- In Italy*, financial statements typically become publicly available within six months after the FYE (stmt_date). To reflect this delay we introduce a 6-month reporting lag.
- After applying the 6-month lag, we define a 12-month forward window, resulting in a full period of: **6 months to 18 months after the statement date**. A default is considered relevant for the model if it falls within this interval.
- We define the target class as follows:

default_flag = 1

(if the default date occurs **within 6 to 18 months** after the stmt_date)

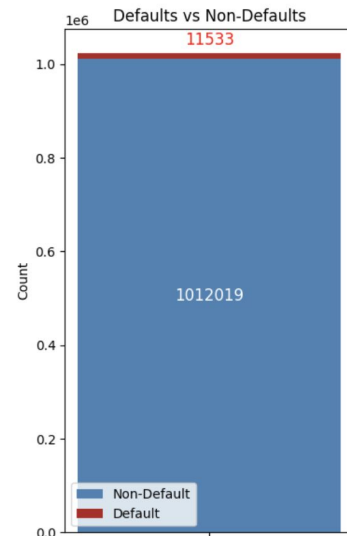
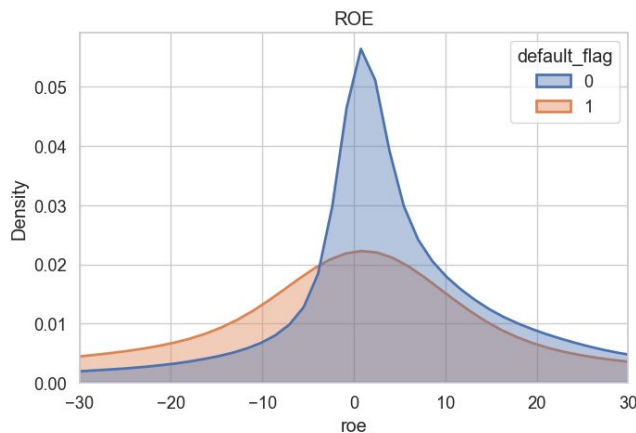
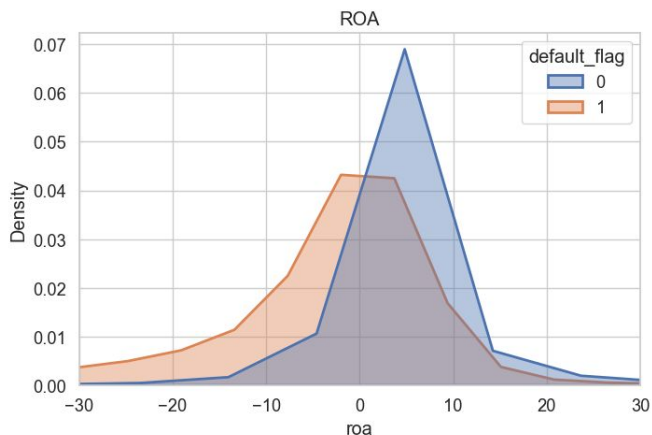
default_flag = 0

(if the default date falls **outside** this window)

*<https://italianbusinessregister.it/en/annual-accounts>

Data Understanding

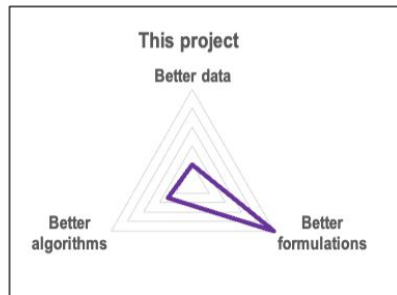
- The dataset contains **1,023,552** observations, **44** features, firm-year records spanning from **2007 to 2012**, and includes companies across **83** unique sectors.
- There are 11,533 defaults firms in the dataset, resulting in a default rate of **1.13%**.
- We dropped the feature: **eqty_corp_family_tot** as it has all Null values and was not useful for our analysis.
- Intuitively relevant features such as ROA and ROE do not separate defaulting from non-defaulting firms in practice, underscoring the need for more rigorous feature construction to extract meaningful default-risk signals.



Ratios

Profitability

- ROA
- ROE
- $\text{EBITDA_Margin} = \text{ebitda} / \text{rev_operating}$
- $\text{EBITDA_to_Assets} = \text{ebitda} / \text{asst_tot}$
- $\text{EBITDA_COGS} = \text{ebitda} / \text{COGS}$
- $\text{Prof_to_COGS} = \text{Profit Operations} / \text{COGS}$
- $\text{Operating_Margin} = \text{prof_operations} / \text{rev_operating}$



Leverage

- $\text{Liability_to_Equity} = \text{liab_tot} / \text{eqty_tot}$
- $\text{Liability_to_Asset} = \text{liab_tot} / \text{asst_tot}$
- $\text{ST_Liab_to_Assets} = \text{liab_st} / \text{asst_tot}$
- $\text{LT_Liab_to_Assets} = \text{liab_lt} / \text{asst_tot}$
- $\text{Debt_to_Equity} = \text{debt_tot} / \text{eqty_tot}$
- $\text{Debt_to_Asset} = \text{debt_tot} / \text{asst_tot}$
- $\text{ST_Debt_to_Assets} = \text{debt_st} / \text{asst_tot}$
- $\text{LT_Debt_to_Assets} = \text{debt_lt} / \text{asst_tot}$

Growth

- Profit_growth
- Revenue_growth
- Asset_growth
(defined later)

Liquidity

- $\text{Current_Ratio} = \text{asst_current} / \text{liab_st}$
- $\text{Quick_Ratio} = (\text{cash_and_equiv} + \text{AR}) / \text{liab_st}$
- $\text{Cash_Asset_Ratio} = \text{cash_and_equiv} / \text{asst_tot}$
- $\text{Cash_Ratio} = \text{cash_and_equiv} / \text{liab_st}$
- $\text{Capital_to_Assets} = \text{wc_net} / \text{asst_tot}$
- $\text{Operating_Cash_Flow_Ratio} = \text{cf_operations} / \text{liab_st}$
- $\text{Fixed_to_Total_Assets} = \text{fixed_asst_tot} / \text{asst_tot}$

Activity

- $\text{Asset_Turnover} = \text{rev_operating} / \text{asst_tot}$
- $\text{Receivables_Turnover} = \text{rev_operating} / \text{AR}$
- $\text{FixedAsset_Turnover} = \text{rev_operating} / \text{fixed_asst_tot}$
- $\text{Cash_Return_on_Assets} = \text{cf_operations} / \text{asst_tot}$

Size

- $\text{Log Assets} = \log(\text{asst_tot})$
- $\text{Log Sales} = \log(\text{rev_operating})$

We could formulate all the above with the data given. But, these are not the final features used.

Why Ratios??

Profitability	A firm that produces a higher return for its investment should have fewer chances of defaulting.
Leverage	A highly leveraged firm is banking more on borrowed money for daily operations. Hence, the chances of it defaulting are higher.
Liquidity	A firm with low liquidity does not have enough assets that can be converted to cash quickly to repay its short-term debt obligations. So, a firm with poor liquidity has higher chances of defaulting on its short-term obligations.
Activity	This measures how efficient a firm is in generating revenue using its assets. An inefficient firm in this case would be more likely to default.
Size	Very small firms would have less diversified revenue streams and limited capital. They are also more likely to be leveraged. This would make them more likely to default.
Growth	A firm is less likely to default if it has positive growth rate.

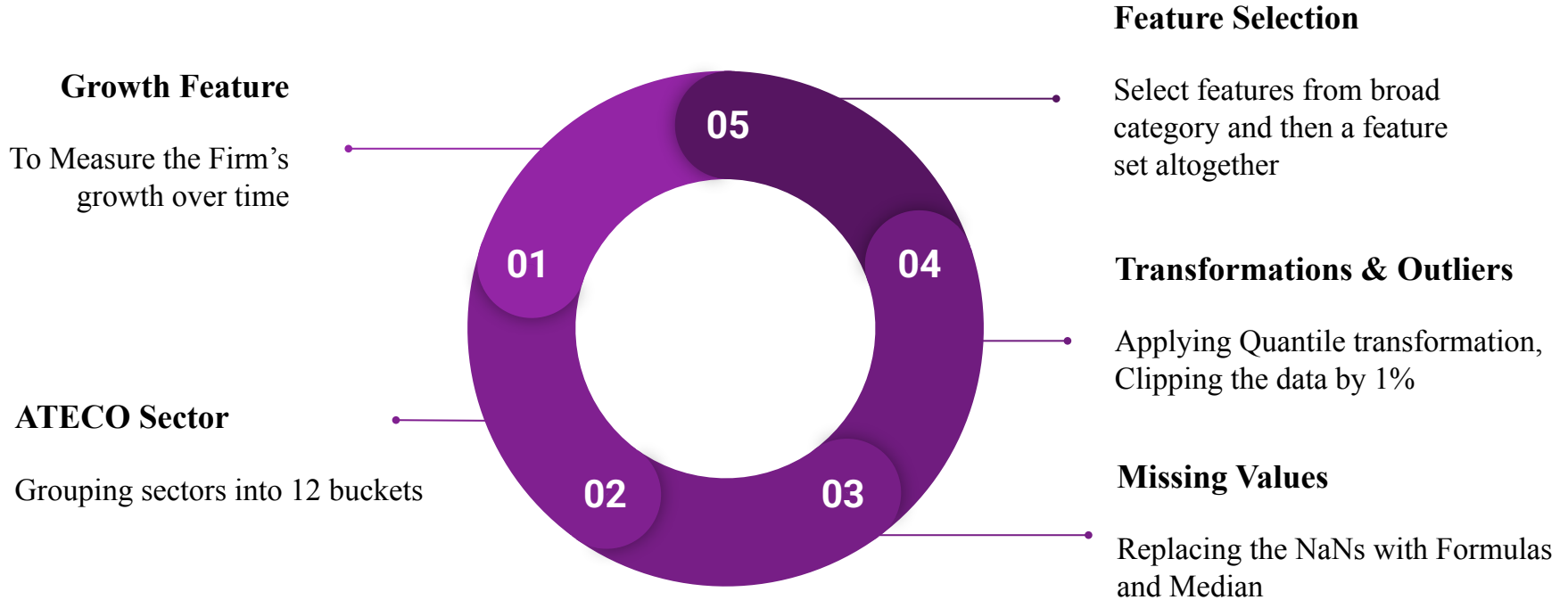
Wait....Wrong Data !!

- In our training data, we found:
 - 1018 observations where $\text{debt} > \text{liabilities}$

In such cases, we have dropped the rows!



Data Preparation



Outlined above are the steps we considered for our data preprocessing.
We will explain each segment and the reasoning behind it in the following slides.

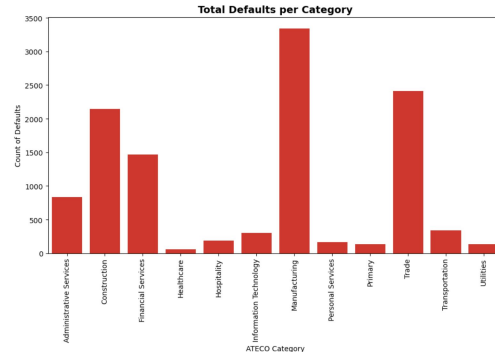
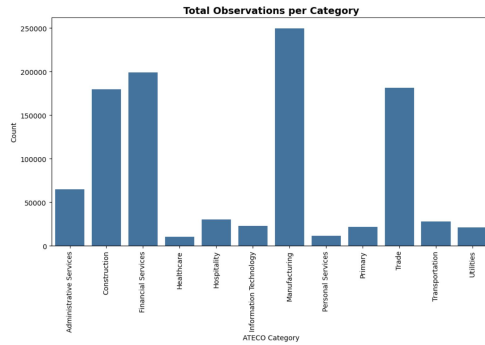
Growth over Time

- We have considered three measures for growth: Assets, Profit, and Revenue.
- To calculate growth rate for training set:
 - First sort the firms by their ID (id) and fiscal year (year of the financial statement)
 - Compute the prior year values (asset/revenue/profit) for each firm
 - For each firm, formulate: **asset_growth** = $(\text{asst_tot} - \text{asst_tot_prev}) / \text{asst_tot_prev}$.
(The same is done for revenue and profit)
 - Special cases:
 - For the firm's first year, growth is set to 0.
 - If prior-year data for a feature is missing or zero, growth is set to NaN to avoid division errors.
- We save the latest growth value for each firm in a CSV file for use in hold-out samples.

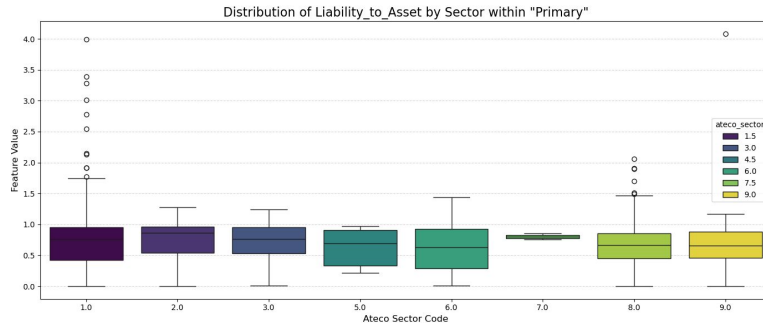


ATECO Sector

- The 83 unique ATECO sectors are divided into 12 major buckets*
- The sector-wise distribution of the Liability-to-Asset ratio within the Primary category is shown below. The similarity of the distributions suggests that aggregating these sectors and choosing the median as a replacement is appropriate.



```
ATECO_CATEGORIES = {  
    "Primary": [(1,3), (5,9)],  
    "Manufacturing": [(10,33)],  
    "Utilities": [(35,35), (36,39)],  
    "Construction": [(41,43)],  
    "Trade": [(45,47)],  
    "Transportation": [(49,53)],  
    "Hospitality": [(55,56)],  
    "Information Technology": [(58,63)],  
    "Financial Services": [(64,66), (68,68)],  
    "Administrative Services": [(69,75), (77,82)],  
    "Healthcare": [(84,84), (85,85), (86,88)],  
    "Personal Services": [(90,93), (94,96), (97,98), (99,99)]  
}
```

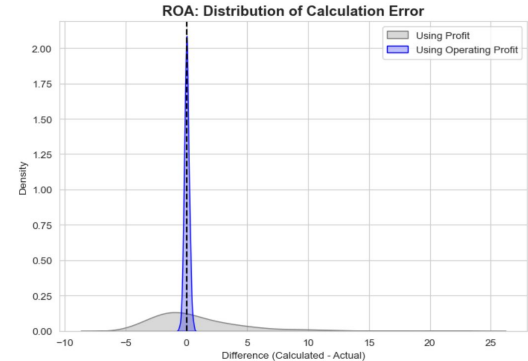
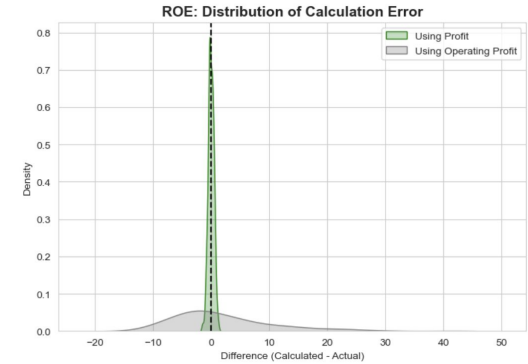


*<https://www.istat.it/classificazione/classificazione-delle-attivita-economiche-ateco/>

Missing Values

Step 1: Replacing NaNs with Economically Motivated Imputation

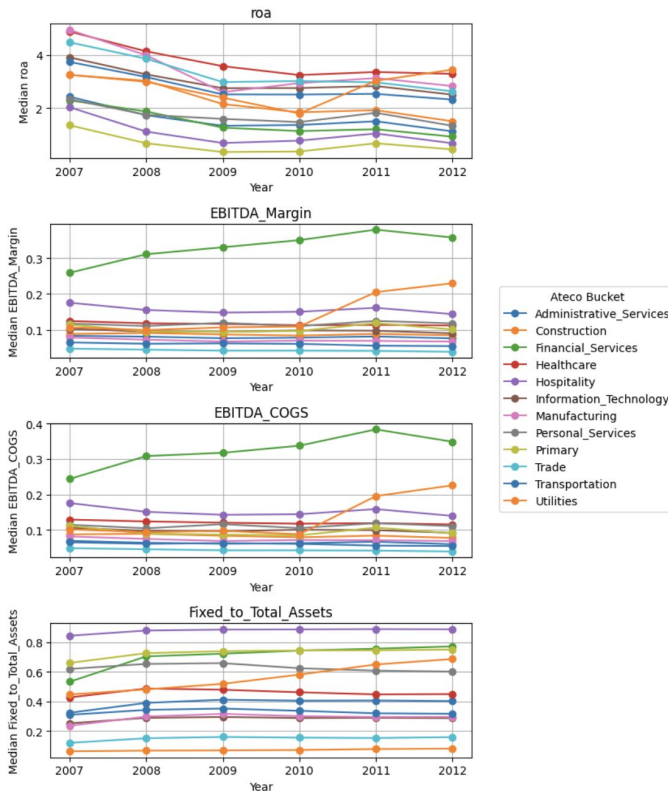
- For example, ROA indicates how profitable a company is in relation to its total assets. Using the indicators of profitability available to us, our analysis indicate that $(\text{prof_operations} * 100) / \text{asst_tot}$, while not the standard definition, served as the most accurate proxy for ROA.
- Similarly $(\text{profit} * 100) / \text{eqty_tot}$, while not the standard definition, served as the most accurate proxy for ROE in this dataset.
- Following accounting rules were used to generate new features and fill in missing values:
 - Total Debt = Short Term Debt + Long Term Debt
 - Total Liabilities = Total Assets - Total Equity
 - St Liabilities = Current Assets - Net Working Capital
 - Fixed Assets = Intangible + Tangible + Financial Assets
 - Operating Revenue = Operating Profit + COGS
 - ROA = Operating Profit / Total Assets
 - ROE = Net Profit / Total Equity



Missing Values

Step 2: Replacing NaNs with Median

- For the remaining missing values after Step 1, we replaced them with median values (calculated w.r.t each year and the 12 ateco categories).
 - Example: For missing EBITDA_Margin values in a firm's 2013 statement, the median value of that firm's ateco category in 2012 will be used as the imputed value.
 - For the hold out sample, we use the most recent available year's median for that category.
 - Our temporal analysis showed three consistent patterns in category-level medians:
 - many features exhibit monotonically increasing category medians over time,
 - others show monotonically decreasing medians,
 - and some are relatively constant year-to-year.
- In all cases, the category median for year t is most strongly correlated with the median for year $t-1$, making it the best forward-looking estimate that avoids look-ahead bias.
- Although category-level median imputation will have high error if the firm in question is an outlier or far away from the median, their low frequency makes this imputation strategy reasonable for a majority of the firms.



Outliers & Transformation

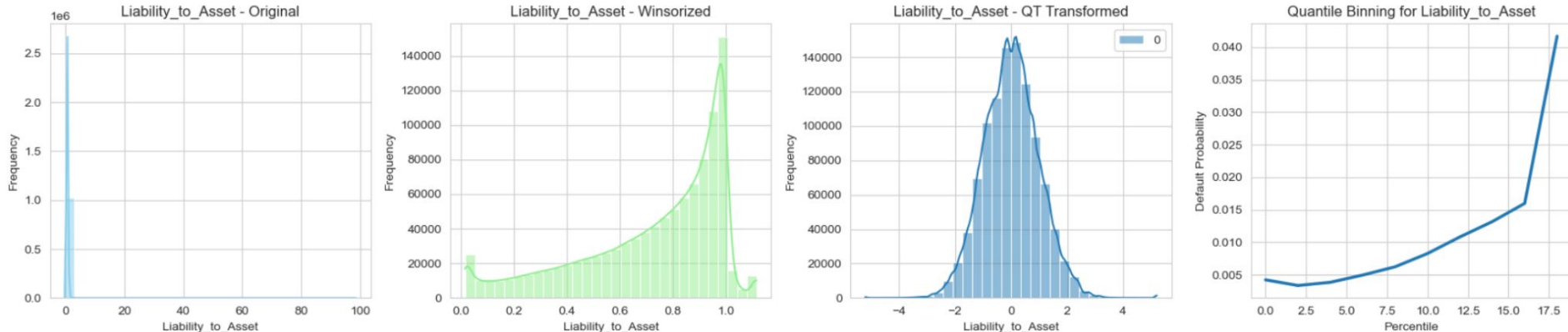
Step 1: We plot the raw column values to visualize the data distribution.

Step 2: We perform 1% winsorization to get rid of any outliers

Step 3: We perform a non-parametric (quantile) transformation on this data.

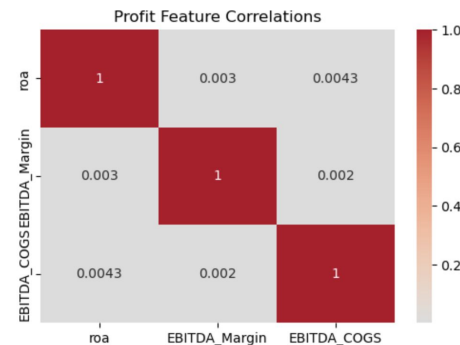
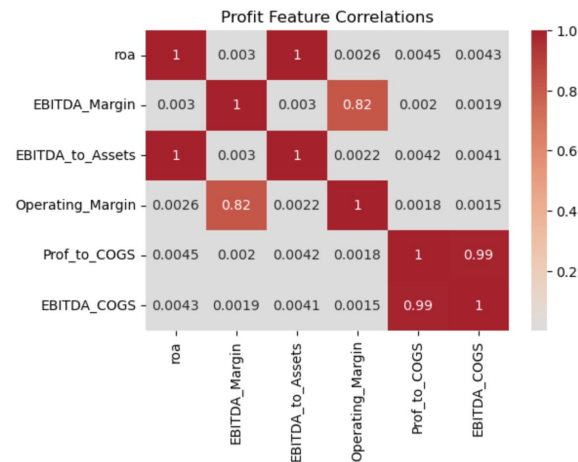
Step 4: We bin the column data and plot with default rate to visualize the relationship between the feature and default probability.

We show the below plots for Liability_to_Asset. The 4th plot highlights an important pattern. As a firm becomes more leveraged, it's probability of defaulting increases.



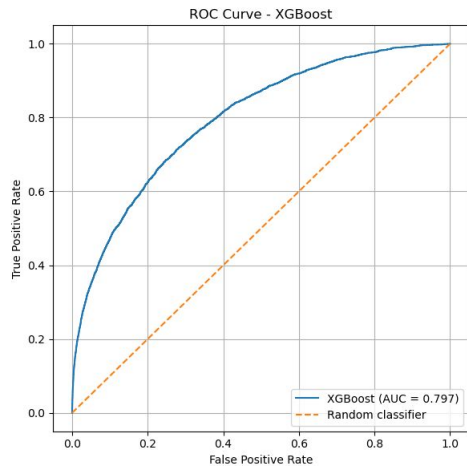
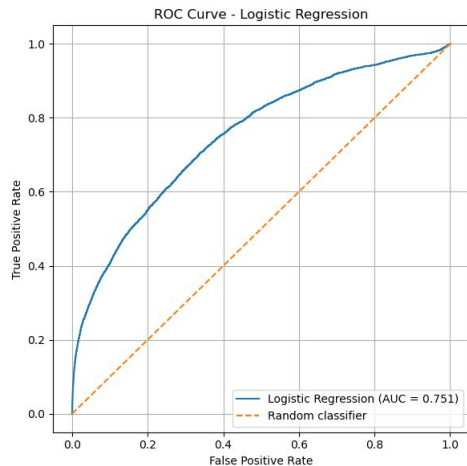
Modeling - Feature Selection

- To construct the final feature set, we first examined the pairwise correlations within each broad feature category. For example, in the *profitability* category, we found that 4 out of 6 variables were highly correlated, indicating redundancy.
- We then evaluated each feature using additional diagnostic criteria, including:
 - univariate logit performance
 - correlation with the default flag
- Based on these combined valuations, we selected a subset of features that were both informative and minimally redundant—for profitability, this led us to choose ROA, EBITDA_Margin, and EBITDA_COGS.
- After selecting features from each category, we perform Variance Inflation Factor (VIF) on the feature set to check for multicollinearity (results in Appendix).
- Our final features: `roa`, `EBITDA_Margin`, `EBITDA_COGS`, `Liability_to_Asset`, `Cash_Ratio`, `Operating_Cash_Flow_Ratio`, `Fixed_to_Total_Assets`, `Receivables_Turnover`, `Cash_Return_on_Assets`, `asset_growth`, `revenue_growth`



Modeling - Logit & eXtreme Gradient Boosting

- We generate our main test dataset in such a way to maintain a similar default rate in both the train (713,485 observations) and test samples (305,779 observations).
- Our baseline model is a Logistic Regression trained on the selected features. It achieves an AUC of **0.7506** on the test dataset.
- We also experiment with an XGBoost model with the same selected features which achieves an AUC of **0.7970**.
- However, on removal of influential observations found using Cook's distance, we find our Logit model to be more robust in its performance as compared to XGBoost, and is hence used for our final submission.
- We further calibrate the model output probabilities to better match the actual default rate present in the dataset.

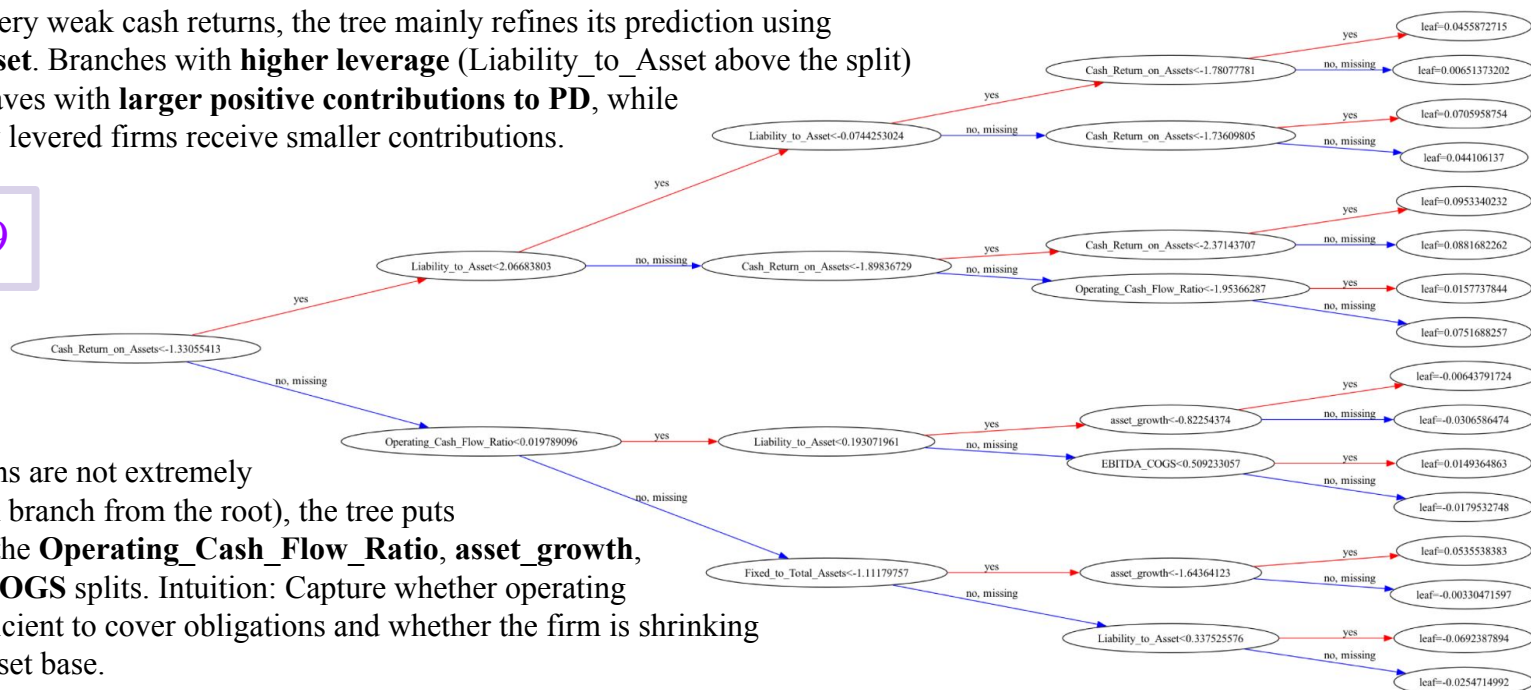


Evaluation - eXtreme Gradient Boosting

The below image is the very first tree (of 100) of our XGB model. Analyzing it provides us some insights on how our model is making the decisions.

- The root node splits on **Cash_Return_on_Assets** < -1.33 => Cash-based profitability is the first thing the model looks at when forming a default prediction.
- Conditional on very weak cash returns, the tree mainly refines its prediction using **Liability_to_Asset**. Branches with **higher leverage** (Liability_to_Asset above the split) tend to end in leaves with **larger positive contributions to PD**, while more moderately levered firms receive smaller contributions.

AUC: 0.79



- When cash returns are not extremely negative (bottom branch from the root), the tree puts more weight on the **Operating_Cash_Flow_Ratio**, **asset_growth**, and **EBITDA_COGS** splits. Intuition: Capture whether operating cash flow is sufficient to cover obligations and whether the firm is shrinking or growing its asset base.

Evaluation - Logit

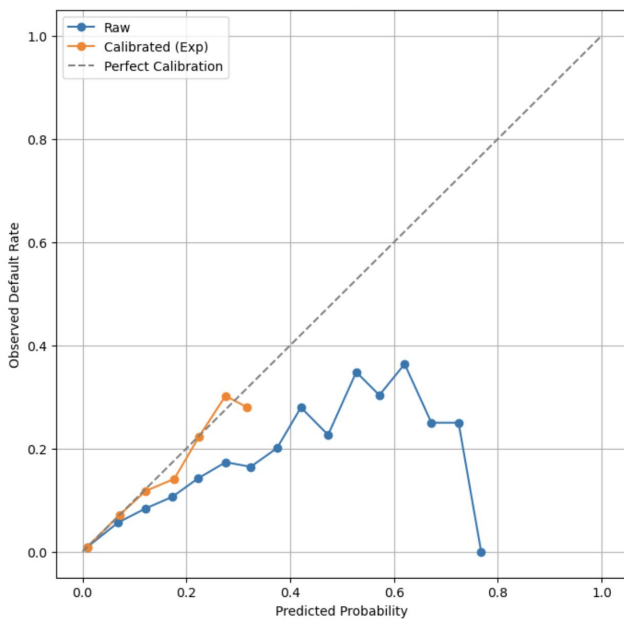
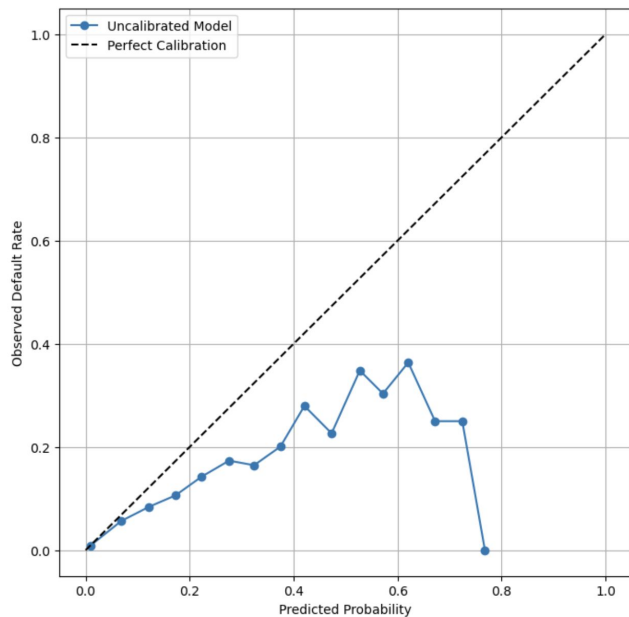
- The model assigns **Liability_to_Asset** (+0.37) as the dominant predictor of default. As a company's leverage increases relative to its assets, the probability of default rises significantly.
- **Cash_Return_on_Assets** (-0.21), **ROA** (-0.16) and **Asset_Growth** (-0.15) serve as the strongest factors against default. High liquidity generation and operational efficiency substantially lower the predicted default risk.
- **Turnover** and **Margin** metrics also possess a moderate negative coefficient, reinforcing that operational health is a key indicator against default.

AUC: 0.75

Feature	Coefficient
Liability_to_Asset	0.373957
EBITDA_COGS	0.001059
Receivables_Turnover	-0.038623
revenue_growth	-0.039989
Operating_Cash_Flow_Ratio	-0.074866
Fixed_to_Total_Assets	-0.076673
EBITDA_Margin	-0.138574
asset_growth	-0.151074
roa	-0.161565
Cash_Return_on_Assets	-0.210240

Calibration

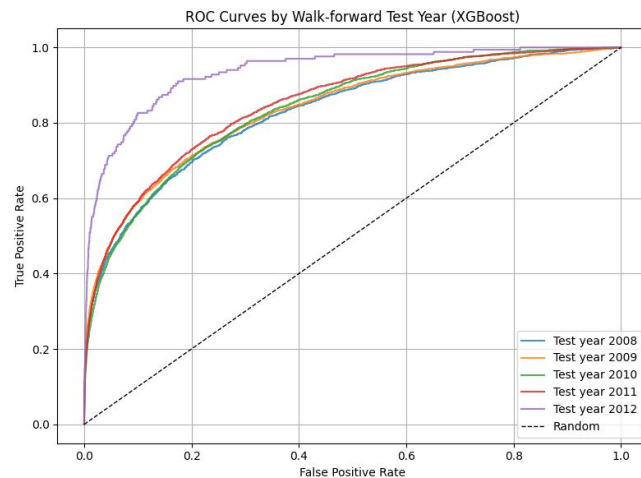
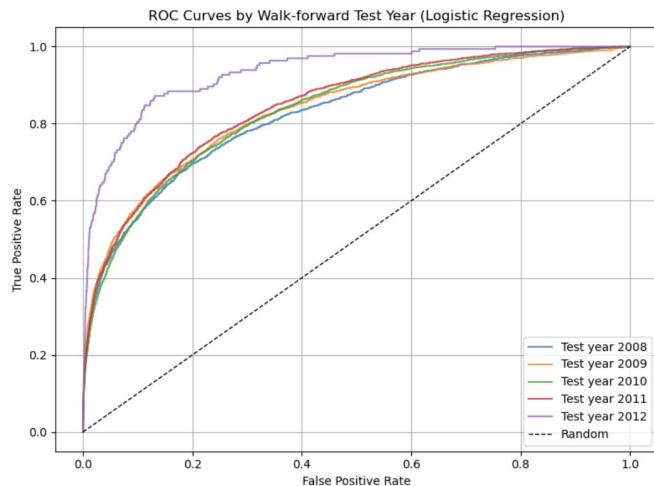
The original calibration curve (left) shows severe underestimation from the model. We fit the predicted probabilities to an exponential non-linear curve to calibrate the predicted probabilities with actual probabilities. The calibrated model (right) shows better alignment with actual probabilities in the low default regions. However, we do not see any values above 0.3.



Walk Forward Analysis

- Both models show a **consistent upward trend in AUC** as additional historical years are added to the training window.
- In each walk-forward iteration, XGBoost outperforms the logistic model: Example: in 2010, XGBoost AUC = **0.838**, while Logit AUC = **0.810**.
- When the training window expands to include **2007–2011**, both models experience a **dramatic increase in test AUC** (although this might primarily be due to the lesser number of defaults in 2012).
- Our logit model's performance is on par with XGBoost on almost all years, and combined with it being better to calibrate, formed our decision to consider it as our final model.

Test Year	Logit	XGBoost
2008	0.8269	0.8281
2009	0.8206	0.8359
2010	0.8107	0.8381
2011	0.8231	0.8499
2012	0.9256	0.9371



Deployment

When an Italian firm publishes its annual financial statements (typically by July), the bank can feed these accounting ratios through our logistic regression PD model, which outputs a calibrated probability of default for the next 12 months.

Potential Issues:

- The logistic model captures statistical associations found in historical Italian defaults. It should not be interpreted as implying that changing a financial ratio will *cause* a change in the PD.
- Some industries in Italy (e.g., manufacturing) are overrepresented, while others are sparse. This may bias PDs when applied to sectors with limited historical representation.

Settings where this model should not be applied:

- To predict PD for non-Italian, financial/insurance or very small firms (< €1.5MM in assets).
- To predict PD based on quarterly financial statements.
- Periods of economic stress in the Italian economy.

Ethical/Legal Considerations: There are regulations like GDPR in the European Union, under which banks must ensure lawful basis for processing financial statements and proper documentation of model use.

Our final harness works by saving the following steps used in preprocessing:

- The final logit model is stored as a joblib file
- Growth(latest) and median values to be considered are stored as a CSV file
- When the harness function is called, our preprocessing functions operate on the provided input
- An output as a CSV file is generated containing calibrated probabilities
- Dependencies: Python, Pandas, NumPy, Scikit-Learn, Joblib

Appendix:

Final Feature	VIF
ROA	1.034850
EBITDA_Margin	1.000414
EBITDA_COGS	1.002790
Liability_to_Asset	1.548750
Cash_Ratio	1.010615
Operating_Cash_Flow_Ratio	1.010626
Fixed_to_Total_Assets	1.551521
Receivables_Turnover	1.000085
Cash_Return_on_Assets	1.036105
asset_growth	1.005051
revenue_growth	1.000273

References

- [1] Act to Credit and Portfolio Management in Practice, Jeffrey R. Bohn, Roger M. Stein
- [2] Beaver, William H. "Financial ratios as predictors of failure." *Journal of accounting research* (1966): 71-111.
- [3] Altman, Edward I. "Financial ratios, discriminant analysis and the prediction of corporate bankruptcy." *The journal of finance* 23, no. 4 (1968): 589-609.
- [4] Ohlson, James A. "Financial ratios and the probabilistic prediction of bankruptcy." *Journal of accounting research* (1980): 109-131.
- [5] Chang, Yung-Chia, Kuei-Hu Chang, and Guan-Jhih Wu. "Application of eXtreme gradient boosting trees in the construction of credit risk assessment models for financial institutions." *Applied Soft Computing* 73 (2018): 914-920.
- [6] Zięba, Maciej, Sebastian K. Tomczak, and Jakub M. Tomczak. "Ensemble boosted trees with synthetic features generation in application to bankruptcy prediction." *Expert systems with applications* 58 (2016): 93-101.
- [7] Ateco Sector Classifications, <https://www.istat.it/classificazione/classificazione-delle-attivita-economiche-ateco/>
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Thank You
